

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q13-0219

Tat Momoli 138 kV
February 26, 2015

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q13-0219. The Interconnection Customer proposes a 164.1 MW natural gas combined cycle facility interconnecting at the Tat Momoli 138 kV substation in the West Virginia area. The request was received on March 28, 2013. The study evaluated system intact and single contingency conditions using the regional planning case, including 136 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 7 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$17,487,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

The GREENHORN - EGAN 138.0 kV Line is overloaded to 124.3% of its 208 MVA emergency rating for the loss of the ROGUE RIV - WEST FORK 345.0 kV L. The project contributes 26.5% DFAX (43.5 MW) to this overload.

For the loss of ROGUE RIVER SUBSTATION - WEST FOK 345.0 kV, the TAT MOMOLI - EGAN 138.0 kV loads to 119.5% of its 239 MVA rating. Project DFAX: 26.5%.

For the loss of ROGUE RIVER - W FORK 345.0 kV line, the GREENHORN - PRUNTYTOWN 138.0 kV loads to 117.0% of its 199 MVA rating. Project DFAX: 26.5%.

The FT MARTIN - LONGVIEW POWER PLANT 138.0 kV is overloaded to 111.6% of its 224 MVA emergency rating for the loss of the AMERICAN RIV - STATE CAMPUS 138.0 kV. The project contributes 5.9% DFAX (9.7 MW) to this overload.

The GREEN BLUFF - GOLD BEACH 138.0 kV Line is overloaded to 105.5% of its 273 MVA emergency rating for the loss of the GREENLAND GAP - SANTEE COOPER 345.0 kV. The project contributes 6.8% DFAX (11.1 MW) to this overload.

The KNOLL TRAIL - PRUNTYTOWN 138.0 kV Line is overloaded to 102.9% of its 217 MVA emergency rating for the loss of the ROGUE RIVER - WEST FORK 345.0 kV. The project contributes 26.5% DFAX (43.5 MW) to this overload.

For the loss of STATE CAMPUS SUBSTATION - AMERICAN RIVER 138.0 kV L, the PIT 7 TAP - FORT MARTIN 138.0 kV L loads to 100.3% of its 237 MVA rating. Project DFAX: 5.9%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
EGAN - GREENHORN 138.0 kV line	Rebuild line with higher-capacity conductor	\$4,652,000	100% (\$4,652,000)
EGAN - TAT MOMOLI 138.0 kV L	Rebuild line with higher-capacity conductor	\$4,215,000	100% (\$4,215,000)
GREENHORN - PRUNTYTOWN 138.0 kV	Rebuild line with higher-capacity conductor	\$5,929,000	100% (\$5,929,000)
FT MARTIN - LONGVIEW POWER PLANT 138.0 kV	Rebuild line with higher-capacity conductor	\$539,000	4% (\$20,000)
GREEN BLUFF - GOLD BEACH 138.0 kV	Reconductor line	\$1,267,000	44% (\$564,000)
KNOLL TRAIL - PRUNTYTOWN 138.0 kV Line	Reconductor line	\$2,064,000	100% (\$2,064,000)
FT MARTIN SUBSTATION - PIT 7 TAP 138.0 kV line	Reconductor line	\$1,155,000	4% (\$43,000)
	Total allocated to project		\$17,487,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.